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# NUMERICAL AND DEVELOPMENT APPENDIX TO NEURAL BELLMAN OPERATORS

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This appendix retains the preceding economic inference, safeguarded selection, localization, certified-readout, deployment and empirical sections without textual deletion. It adds the full new query-reuse and rational-integration records and states the independent-verification boundary. The proofs are in the Technical Supplement; the original broader applications remain in their unchanged companion editions.

## APPENDIX A: FROM RETURNED POLICIES TO ECONOMIC RESOURCE DECISIONS

An all-state upper bound and a comparison of actual policy costs answer different economic questions. We now connect the policies returned by a construction frontier to direct cost inference and then to a choice of implementation resources. The construction data select the policies; independently generated evaluation paths assess them. This separation permits a valid comparison of first-crossing policies with different resolutions.

### A.1. *Simultaneous inference after construction*

Let $\mathcal{H}$ denote all construction information: primitive inputs, numerical labels, fitted continuations, returned policies and complete certificate ladders. Conditional on $\mathcal{H}$, fix a finite catalogue of implemented-policy contrasts. Each implementation includes its observation rule, original witness or nearest-node selector, true-state feasible repair and action quantization. For a contrast $i$, let $D_i$ be the difference in expected discounted costs under a specified initial law. A common-innovation path score evaluates the two policies on their own endogenous trajectories. It is not the difference between two regret bounds.

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Suppose the initial and innovation uniforms are partitioned into equal bins. Independent uniform bin indices are drawn, and interval simulation encloses the score for every continuous realization inside the drawn bins. All unresolved state, selector and repair branches remain in the enclosure. If $[A_i, B_i]$ is a deterministic support interval, clip the path endpoints to that support. Write $\underline{m}_i, \overline{m}_i$ for outward sample-mean bounds on the lower and upper endpoint samples and $v_{i,L}, v_{i,U}$ for upper bounds on their unbiased sample variances. For $n \geq 2$ define

$$R_i(v) = \sqrt{2v\ell/n} + \frac{7(B_i - A_i)\ell}{3(n-1)}, \qquad \ell \geq \log(4K/\alpha), \tag{1}$$

where $K$ bounds the number of sampled contrasts.

THEOREM 1—Construction-conditional economic intervals: *Under the stated independent-bin model, conditional on any valid construction history $\mathcal{H}$, with probability at least $1 - \alpha$ all catalogue contrasts satisfy*

$$D_i \in [l_i, u_i] := [\max\{A_i, \underline{m}_i - R_i(v_{i,L})\}, \min\{B_i, \overline{m}_i + R_i(v_{i,U})\}]. \tag{2}$$

*The same coverage holds unconditionally. Subsequent selection among these covered contrasts, or evaluation at any deterministic implementation-price vector, needs no additional union bound. This conclusion applies only to policies and initial laws included in the catalogue; it does not create direct-cost observations for every unstudied tolerance or initial state.*

The proof is in Supplement, Section 2.10.

In particular, selecting the first policy to meet a specified certificate is permitted when that selection uses construction records rather than the subsequent evaluation paths. Repeated construction clocks are not independent draws from a neural-optimizer population. The statistical randomness here belongs to the evaluation design. A fixed pseudorandom stream implements a reproducible experiment; it is not itself a proof of the independent-bin assumption.

### A.2. *Replacement-fee identification*

Write $D = J^W - J^F$. If witness is installed, replacing it by FVI changes total cost by $k - D$, where $k \geq 0$ is the replacement fee. If FVI is installed, the change is $k + D$. Thus the two directional gains before fees are $D$ and $-D$.

COROLLARY 2—The break-even fee frontier: *On an event where $D \in [l, u]$, the two directional gains belong to $[l, u]$ and $[-u, -l]$. The unknown absolute cost difference satisfies*

$$|D| \in [\text{dist}(0, [l, u]), \max\{|l|, |u|\}]. \tag{3}$$

*For every $k \geq \max\{0, u, -l\}$ neither replacement strictly lowers expected cost. By contrast, $k \geq \max\{0, l, -u\}$ only removes a* certified *strictly profitable replacement. The two thresholds generally differ. Strict inequalities in the first criterion certify that both replacements strictly increase cost.*

The proof is in Supplement, Section 2.11.

The fee frontier is more informative than a conclusion at one generous charge. The earlier $1/64$ replacement decision is retained with its original interpretation and confidence account. The new analysis reports the two directional gain intervals and both thresholds instead of selecting a favorable fee after seeing the data. There is no welfare calibration hidden in the units: a decision maker must supply the economically relevant charge, or use the full frontier to see what remains unidentified.

### A.3. *Observation costs on the same bit grid*

Let $\mathcal{B} = \{4, \ldots, 16\}$ be the available bit precisions. For one fixed returned controller, write $J_b$ for the expected cost of its actual $b$-bit, robustly repaired and quantized implementation. Use $b_0 = 16$ as a common reference and obtain simultaneous intervals for $D_b = J_b - J_{b_0}$. The identity interval at $b = b_0$ is exactly $[0, 0]$. Let $s_b = db \sum_{t<T} B_t$ be the discounted number of coordinate-bits purchased; the net cost is $J_b + \lambda s_b$.

THEOREM 3—Certified choice of actual net implementation cost: *Suppose all* $D_b \in [l_b, u_b]$ *simultaneously. For any* $\lambda \geq 0$ *define*

$$L(\lambda) = \min_b \{l_b + \lambda s_b\}, \qquad U(\lambda) = \min_b \{u_b + \lambda s_b\}, \tag{4}$$

$$\widehat{b}(\lambda) = \min \arg\min_b \{u_b + \lambda s_b\}.$$

*Then, simultaneously for all* $\lambda \geq 0$,

$$0 \leq J_{\widehat{b}} + \lambda s_{\widehat{b}} - \min_{b \in \mathcal{B}} (J_b + \lambda s_b) \leq U(\lambda) - L(\lambda). \tag{5}$$

*The lower and upper functions are finite piecewise-affine envelopes. All changes in their active lines occur at nonnegative intersections of catalogue lines, with exact comparisons deciding ties. The smallest selected precision* $\widehat{b}(\lambda)$ *is weakly decreasing in* $\lambda$.

The proof is in Supplement, Section 2.12.

This result makes a statement about *actual* net cost within the declared bit catalogue. It differs from minimizing a uniform all-state loss bound. Both criteria are useful: the latter gives distribution-free protection relative to the full-information optimum, while (5) uses a specified initial law to compare feasible implementations of one returned policy. Neither criterion gives an optimum over arbitrary sensor technologies or all policies.

For a fair comparison the conventional actor receives its own acquisition allowance. Suppose its nearest-node policy has a certified selected-policy account at a reported state, obtained from the graph modulus $L_t^{\mathrm{gr}}$, and its actual interpolated critic has modulus $L_t^f$. Transporting the same selected node action to the true state, then using cell-wide capacity repair, adds at most

$$(L_t^{\mathrm{gr}} + L_t^f + 2D_t K_t^A) r(b) + D_t \Delta_t. \tag{6}$$

Indeed, transport the feasible node pair directly to the true state, bound its node distance by the reported-state distance plus $r(b)$, and bound $f_t(\widehat{x}) - f_t(x)$ by $L_t^f r(b)$. Robust repair adds $D_t(2K_t^A r(b) + \Delta_t)$. This proves (6) without assuming actor continuity. For a witness critic $L_t^f = L_t^{\mathrm{gr}}$, it reduces to the preceding witness allowance. The empirical comparison therefore evaluates the all-state-account minimizer, the

interval upper-net-cost selector, and a descriptive midpoint selector on the *same* integer bit and price grid. The midpoint selector is not called the true optimum.

## APPENDIX B: CERTIFICATE-GATED NEURAL QUERIES

A prediction of the optimal action and a prediction of a useful numerical query are different objects. Inserting a learned action before examining an already available certificate charges an extra continuation query even when the two boundary actions suffice. We instead make the demand for another query a mathematical decision and let a trained NBO selector determine its location only after that demand has been established. The underlying continuation recursion and the economic optimum are unchanged.

At a recursive state, start with the two feasible action endpoints. Compute the global lower bound from the parabolic minorants of Lemma 6, including the capacity remainder, and the smallest upper endpoint. Return immediately when their difference is at most the requested tolerance. Otherwise choose a partition interval with smallest current minorant. A learned action is used only if it lies in the central half of that interval; it then replaces the conventional proposed split. Outside that region the safeguarded parabolic split is used. The rounded split must pass the one-fifth interior check before evaluation. A prediction is evaluated at most once at each recursive state and is reused only as a proposed location. It supplies neither a value endpoint nor a lower bound.

THEOREM 4—Gated prediction and complete-policy validity: *Under the assumptions of Theorem 7, the gated service returns the same original-optimum and actual-policy guarantees for every supplied prediction, independently of fitting error. In real arithmetic, with endpoint-plus-capacity width at most* $5\delta/8$*, a sufficient number of endpoint evaluations at a state is*

$$A_r(\delta, x) = 4 + \left\lceil 5c(x)\sqrt{H_r/(3\delta)} \right\rceil .$$

*Each requested evaluation either reduces the unresolved action partition or terminates the query; no initial interior evaluation is mandatory. If the boundary certificate suffices, no predictor is evaluated at that state. If every prediction is declined at every recursive*

*query, the value endpoints and implemented actions coincide with those of the conventional parabolic service using the same arithmetic and tie rules.*

The proof is in Supplement, Section 2.8.

This theorem establishes robustness to the numerical content of a fitted prediction, not that learning is free or always useful. It also does not state that an arbitrary routed transcript has fewer queries than the adaptive conventional transcript. The executed comparison records both possibilities. Unlike a policy catalogue with a stored reference action at every state, neither service constructs a conventional state lattice before it can return a certified policy.

### B.1. *State dimension, working memory, and reliability*

For the original scalar model, the primitive bounds $dG_r \le 27$ and $dM_r \le 54$ imply

$$H_r \le \frac{3369}{256}, \qquad \Lambda_r \le \frac{1319}{128}, \qquad \frac{\beta M_{r-1} d}{6144} \le \frac{135}{16384}.$$

Thus the action and scalar-innovation budgets at fixed remaining horizon and tolerance are independent of the number of state coordinates. Every primitive evaluation still requires $O(d)$ arithmetic. A width-$w$ one-hidden-layer ReLU prediction on the recorded $d+5$ features requires $O(dw)$ work and stored parameters; a full quadratic prediction on these features uses $O(d^2)$ terms. Neither cost is included for free in the Bellman recursion.

Let $q_r(\delta)$ be the dyadic quadrature count and let $P_r(d)$ be the cost of one predictor evaluation. Apart from the separately identified exact-arithmetic recovery, a conservative work recursion is

$$W_r(d,\delta) \le CdA_r(\delta) + P_r(d) + A_r(\delta) q_r(\delta) W_{r-1}(d, \delta/(4\beta)).$$

The predictor term is absent on a boundary-certified query. This expression removes the exponential state lattice, not the horizon or innovation-tree dependence. A depth-first scalar implementation can retain the current action partition and one innovation child at a time. The executed batched implementation instead reports its maximum state batch and process peak memory. A zero count of stored state-lattice nodes does not mean zero storage.

The deterministic certificate is valid for any supplied finite network weights. A failure to fit and return such weights is a separate service event. Our reliability estimand is the return count and worst measured complete work over the preregistered finite seed catalogue, with every repetition retained. Numerical workload paths are not optimizer replications. The real-arithmetic cap and the exact-rational precision condition establish algorithmic termination at admissible tolerances; recorded processor seconds are descriptive, not a hardware-uniform worst-case bound.

### B.2. *An explicit economic purchase criterion*

The decision maker buys a callable controller rather than a table of successful sample decisions. Fix a physical capital numeraire for the normalized state, a conversion $m > 0$ dollars per unit of the model's loss index, a computation charge $p \geq 0$ dollars per measured second, and a fixed deployment fee $f \geq 0$ dollars. For $n$ uses under a specified initial-state law, the monetary objective is

$$\mathcal{C}(\pi) = nm\,\mathbb{E}[J_0^\pi(X_0)] + pW(\pi) + f.$$

These are declared preference and service-price parameters, not estimated economic primitives. The policy theorem bounds the expected loss under the original continuous law for every such initial-state law. The finite dyadic workload below measures execution; its average cost is not substituted for that expectation.

If two policies have an independently valid expected-gain interval $[l, u]$ for $\mathbb{E}[J_0^{\pi_0} - J_0^{\pi_1}]$, and their incremental measured service charge is $\Delta = p(W_1 - W_0) + f_1 - f_0$, then the monetary net gain lies in

$$[nml - \Delta,\ nmu - \Delta].$$

This follows by multiplication by the positive numeraire conversion and subtraction of the same service charge. A positive lower endpoint supports purchase; a negative upper endpoint rules it out at the stated prices. Uniform regret certificates alone do not supply a directional expected-gain interval. They instead support a worst-case accuracy requirement imposed before selecting the least expensive eligible service.

Finally, a forecast of amortized query work is distinguishable from a measured complete service. With a fixed model and exactly repeated query workload, if training requires $B$ operations and deployment requires $D_L$ rather than $D_C$ operations per workload, then learning uses fewer total operations after $k$ repetitions precisely when $D_L < D_C$ and $k > B/(D_C - D_L)$. If $D_L \geq D_C$ and $B \geq 0$, no repetition of that unchanged workload produces an operation saving. This arithmetic criterion includes training and cannot be applied to unrelated future states without a new assumption. The executed catalogue reports its own full process clocks, not a favorable hypothetical amortization.

## APPENDIX C: WHEN A PREDICTED QUERY COMPLETES THE CERTIFICATE

The safeguarded theorem is insensitive to the quality of a supplied prediction. A separate question is when that prediction is sufficient to finish verification. The answer must concern both the selected action and the unresolved lower bound. Proximity to a minimizer alone does not make a coarse lower certificate accurate.

PROPOSITION 5—Witness accuracy and certificate localization: *Let $f$ be a differentiable, $H$-semiconcave objective with $H \geq 0$ on $[0, c]$, let $a^*$ minimize $f$, and suppose $|f'| \leq \Lambda$. An evaluated proposal $\widehat{a}$ lies in $[0, \underline{c}]$, where $0 < \underline{c} \leq c$ and $c - \underline{c} \leq \Delta$. Each endpoint interval is a valid enclosure of $f$ and has width at most $w$. Suppose the partition containing $\widehat{a}$ covers $[0, \underline{c}]$ and has maximum interval length $h$. Let $L$ be its parabolic lower certificate with the capacity allowance $\Lambda\Delta$, and let $U$ be the smallest upper endpoint, retaining earlier valid bounds as in the query algorithm. With $e = |\widehat{a} - a^*|$ and $\lambda = |f'(a^*)|$,*

$$U - L \leq 2w + \Lambda\Delta + \frac{Hh^2}{8} + \lambda e + \frac{He^2}{2}. \tag{7}$$

*For an interior minimizer $\lambda = 0$. At a boundary the linear term cannot generally be removed. Thus the right side at most $\delta$ is a sufficient certificate-return condition, not an assumption that a trained prediction meets it.*

*If old and new valid endpoint records give $(L, U)$ and $(L', U')$ after one additional query, with the running maximum lower bound and minimum upper bound, then*

$$(U - L) - (U' - L') = (U - U') + (L' - L) \geq 0. \tag{8}$$

*An added prediction may have zero certified progress. The same identities apply to a conventional query.*

The proof is in Supplement, Section 2.9.

This proposition isolates what a fitted representation can contribute without crediting it with a conventional solution. It can provide a small-error witness and change which unresolved interval is refined. The global partition, endpoint arithmetic and capacity account still have to support that witness. The maximum-width sufficient condition is deliberately conservative; the implemented minimum-minorant rule can stop earlier when only a subset of the partition remains competitive.

### C.1. *A complete accuracy contract and its resource dependence*

For $T$ dates and target $\epsilon$, write $S_T = \sum_{t<T} \beta^t$. The recorded service chooses a common local tolerance by rounding $\epsilon/(2S_T)$ to binary64 and verifies the final outward allowance. Under $dG_r \leq 27$, a uniform per-date acquisition allowance at forty bits is

$$\alpha_b = \left(8 + \frac{297\beta}{16} + 27\right) 2^{-40}.$$

The service certifies $(\delta + \alpha_b)S_T \leq \epsilon$. Each descendant request is $\delta/(4\beta)$, rounded as in the actual code and included in its outward endpoint guard. At the smallest request in the two- and three-date catalogues, the declared thirty-two-bit action quantum satisfies the exact-recovery condition $\delta_r > 8\Lambda_r 2^{-32}$. The supplement checks this inequality with rationals for every horizon and target, rather than inferring it from a successful workload.

For fixed remaining horizon and accuracy, the bounds on action endpoints and scalar-innovation bins are independent of state dimension in this normalized economy. The factors $dG_r$ and $dM_r$ are bounded, while every transition and cost still reads $d$ coordinates. A width-$w_\theta$ one-layer ReLU predictor has $O(dw_\theta)$ parameters

and evaluation work; a dense quadratic on the same features has $O(d^2)$ terms. These statements identify the absence of a $(n+1)^d$ stored state table. They do not count an arbitrary-precision rational operation as a unit-cost floating operation and do not suppress the product of descendant query counts as horizon grows.

The two fitted implementations and their conventional controls therefore share one original-optimum contract, but need not share a transcript, a returned policy or a runtime. The completed experiments charge their distinct transcripts. A finite-catalogue reliability statement concerns all specified seeds and repetitions, including fitting warnings. It is not a population statement about optimizer success.

## APPENDIX D: MATCHED READOUTS AND COMPLETE RECOVERY

For $x \in [0,1]^d$, the two fixed feature maps used in the comparison are

$$\phi^{\mathrm{R}}_{jh}(x) = \frac{1}{d}(1 - |4x_h - j|)_+, \qquad \phi^{\mathrm{B}}_{jh}(x) = \frac{1}{d}\binom{4}{j} x_h^j (1-x_h)^{4-j}, \quad 0 \le j \le 4,$$

with $1 \le h \le d$. Both are nonnegative partitions of unity with $5d$ box readout coefficients. The first is an additive hat basis with an exact ReLU realization; the second is a degree-four Bernstein basis. Both purchase the same verified boundary contexts and approximate action labels, and optimize the same certificate loss. The label-loss control changes the objective, not its hat features. The achieved rational optimization certificate is evaluated after quantization.

### D.1. *Complete recovery under a validation law*

The validation design draws each coordinate as $U/2^{16}$, with $U$ uniform on $\{0, \dots, 2^{16}-1\}$. These states are representable by the twenty-four-bit policy acquisition contract, but are not uniform over that finer grid. The following result applies to the declared independent context law and recovery. It does not assert that on-policy recursive states have that law.

PROPOSITION 6—A finite catalogue of complete work: *Condition on a frozen finite family of predictors and a predictor-independent validation law. Suppose a failed screen invokes a deterministic recovery whose root-query count is at most $K(z)$, and every*

*root query costs at most $W(z)$ under the declared descendant and arithmetic contract. If $0 \le W(z)K(z) \le C_{\max}$, then with probability at least $1-\alpha$, simultaneously for $M$ fixed predictors,*

$$\mathbb{E}[\textit{recovery work}_\theta] \le \frac{1}{n}\sum_{i=1}^{n} W(z_i)K(z_i)\ell_\tau(z_i,\theta) + C_{\max}\sqrt{\frac{\log(M/\alpha)}{2n}}.$$

*Training, context acquisition, prediction, loading, identity checks and output costs are additional terms in complete work. Replacing $W$ by one gives a root-query result, not a bound on elapsed process time.*

*Proof.* See Supplement, Section 4.1.

The executed validation uses the root-query specialization and records descendant queries separately. It does not fit a runtime model and declare it a valid worst-case endpoint cost. The predictable-deployment result in Section E adds an explicit context-transfer premise and a complete operation account.

### D.2. *An additional vector witness*

In the two-control economy a trained, projected and downward-rounded proposal can supply a new evaluated upper witness. Its insertion leaves the triangular lower cover unchanged. The original corner endpoints remain in that cover.

PROPOSITION 7—Optional vector witnesses: *Suppose a triangular action cover has a valid global lower bound $L$ and feasible evaluated witnesses with upper bound $U$. Adding any finite number of feasible, correctly evaluated witnesses and replacing $U$ by the minimum of all their upper bounds preserves containment of the original feasible infimum. Every returned witness with $U - L \le \delta$ has original Bellman regret at most $\delta$. The conclusion is independent of the training method and of whether an added witness is a vertex of the lower cover.*

*Proof.* See Supplement, Section 4.2. The vector benchmark changes the trained proposal or the choice of a triangle to refine, not the authority of the global lower cover. Its training objective remains action-label loss.

## APPENDIX E: FROM VALIDATED READOUTS TO COMPLETE DEPLOYMENT

The fixed-feature construction and its optimization certificate identify an achieved numerical object. The next issue is its use after training. Independent validation need not imply that the states subsequently visited by a controller are independent. We give a work account that keeps those two sources of information separate and charges the descendants of a failed screen.

### E.1. *Predictable selection and changing context laws*

Condition on a frozen family $\Theta$ of $M$ readouts. Write

$$f_\theta(z) = K(z)\ell_\tau(z,\theta), \qquad 0 \le N_\theta(z) \le f_\theta(z) \le K_{\max},$$

where the fallback and its endpoint cap are the ones specified for the context. Let $Z_1^v, \dots, Z_n^v$ be independent validation contexts with law $P$, independent of readout construction. Set

$$u_\theta = \min\left\{ K_{\max},\ \frac{1}{n}\sum_{i=1}^{n} f_\theta(Z_i^v) + K_{\max}\sqrt{\frac{\log(M/\alpha)}{2n}} \right\}. \tag{9}$$

During deployment, let $\mathcal{F}_{b-1}$ contain construction, validation, and all observations before block $b$. The selected readout $\theta_b$ is $\mathcal{F}_{b-1}$-measurable. The new context $Z_b$ can depend on this history. Suppose a declared predictable number $\rho_b \in [0,1]$ satisfies

$$\|\mathcal{L}(Z_b \mid \mathcal{F}_{b-1}) - P\|_{\mathrm{TV}} \le \rho_b, \qquad \|P - Q\|_{\mathrm{TV}} = \sup_A |P(A) - Q(A)|. \tag{10}$$

This is a distributional condition, not a consequence of matching a model hash.

Use an additive operation model. Let $F_0$ bound fitting, serialization and validation work. Let $h_b$ bound loading, identity checks, prediction, previously purchased endpoints, acquisition and durable output at block $b$. Finally let a nonrandom $c_b$ bound the complete descendant and numerical-recovery work of one additional root endpoint. The budgets must cover all admissible histories, not merely the largest observed runtime. Assume the deployed work satisfies

$$W_B \le F_0 + \sum_{b=1}^{B} \{h_b + c_b N_{\theta_b}(Z_b)\}.$$

This includes a complete-action mesh fallback when that is the fallback validated by the experiment. Substitution of a different fallback requires its own cap and loss contract.

THEOREM 8—Validated work under predictable deployment: *For a fixed deployment window $B$ and $0 < \alpha, \gamma < 1$, with probability at least $1 - \alpha - \gamma$,*

$$W_B \le F_0 + \sum_{b=1}^{B} h_b + \sum_{b=1}^{B} c_b \min\{K_{\max}, u_{\theta_b} + K_{\max}\rho_b\} + K_{\max}\sqrt{\frac{1}{2}\log(1/\gamma)\sum_{b=1}^{B} c_b^2}. \tag{11}$$

*The readout can be selected using the validation sample and the preceding deployment history. The conclusion does not require the deployment contexts to be independent. Under the same local feasibility and certificate premises, this selection does not change the original-optimum policy guarantee.*

*Proof.* See Supplement, Section 3.7.

This theorem gives a prospective route from training to complete work rather than interpreting a training loss as a timing result. A smaller root-query mean is useful when its descendant-weighted savings exceed setup and per-block overhead. The $c_b$ terms expose the continuation-tree cost. Setting them to one would prove a root-query statement, not a full-service statement. The R69 independent-context experiment below uses its declared mesh recovery; its adaptive recursive deployment records are not claimed to satisfy an empirically verified version of (10).

### E.2. *Policy identity, declined reuse, and adoption*

Cost inference can be combined with a deployment clock only after identifying the controller whose cost was estimated. The release checks the complete readout arrays, tolerance, acquisition precision, action rounding and original kernels. Consider a service processing $B$ blocks of $m$ trajectories each. In $D$ predetermined blocks it declines an incompatible readout and uses the benchmark $\nu$; otherwise it uses a fixed $\pi$. Suppose each block has the same initial law and economic primitives, and

the decline schedule is independent of the trajectories. Let $[l, u]$ be a simultaneous interval for $\Delta = \mathbb{E}[J^{\pi} - J^{\nu}]$.

PROPOSITION 9—Complete-service adoption region: *Let $W_{\pi}, W_{\nu}$ be complete work observations under the same boundary. If $v \geq 0$ values one normalized expected-cost unit, $p \geq 0$ prices one work unit, and $f \geq 0$ is the incremental installation fee, then the net cost difference belongs to*

$$[\, vm(B - D)l + p(W_{\pi} - W_{\nu}) + f,\ vm(B - D)u + p(W_{\pi} - W_{\nu}) + f \,] \tag{12}$$

*on the original simultaneous event. The statement holds for every such price triple on that event. With observed work saving $S = W_{\nu} - W_{\pi} > 0$, strict adoption is supported whenever*

$$pS > f + vm(B - D)u.$$

*If work is instead bounded by valid intervals, replacing $W_{\pi} - W_{\nu}$ by its interval difference preserves the statement.*

*Proof.* See Supplement, Section 3.8.

The executed reuse design has $B = 64$, $m = 8$, and one predetermined identity rejection, so its normal-policy multiplier is 504. This does not make an observed prefix clock a completed service, or an observed timing range a confidence interval for future hardware. The paired complete receipts and the conservative envelope across the observed repetitions are both reported. The inputs $v, p, f$ remain declared decision parameters; the experiment estimates neither an installation fee nor the monetary value of normalized investment losses. The policy-cost margin supplies an economically interpretable eligibility condition, while the full adoption region keeps computational savings and policy quality separate.

## APPENDIX F: TRAINING, REUSE, AND POLICY-COST PRECISION

The experiment links the constructive readout to the actual controlled economy. It compares representation and training loss at a common information budget, validates a frozen readout on independent contexts, measures a longer deployment window, and evaluates the returned controllers at a prespecified policy-cost margin.

The original continuous laws, state-dependent investment capacity and Bellman target are unchanged.

### F.1. *Design and information balance*

The scalar tasks are $(d, T, \varepsilon) = (2, 2, 1/128)$ and $(8, 3, 1/64)$, with twenty-four-bit state acquisition and thirty-two-bit downward action rounding. Four fitted proposals receive the same forty-eight training states, approximate action labels and verified boundary endpoints per nonterminal date. The certificate-aware hat–ReLU and degree-four Bernstein readouts have exactly $5d$ box-constrained coefficients and the same objective, penalty and optimizer budget. An action-label readout on the same hat features separates the objective from the representation. A quadratic label fit retains its explicitly different parameter count. Adaptive parabolic search and upper-chord minimization are prediction-free controls; unused training is not charged to them.

The hat basis has an exact finite ReLU realization. It also has a conventional piecewise-linear interpretation, so this comparison does not count two encodings of that basis as independent algorithms. The Bernstein comparison holds the coefficient count and purchased information fixed. All fitted proposals use the same final certificate. Every returned certificate-aware iterate is quantized and receives an exact rational value of the objective and box dual gap from Proposition 10. The achieved gap is reported rather than assuming that an optimizer's success flag proves optimality.

The protocol fixes two fitting seeds and four process repetitions. It contains eighty cold comparisons, sixteen independent-validation services, fifty-six reuse services, two paired-inference services and thirty-four vector services: 188 complete processes. All returned. This is finite-catalogue evidence for the complete verifier, not a population claim about fitting success. Four clock repetitions are not four new training samples. The construction, receipts and original source freeze predate this manuscript integration; the publication replay adds no scientific observations.

The complete cold-start results are reported in Supplement Table L.I. At a fixed purchased-information and coefficient budget, the hat–ReLU readouts close 48 and

50 of 256 small-task validation contexts for the two seeds, compared with 25 and 21 for Bernstein. The achieved exact box dual gaps are at most $10^{-7}$ across the certificate-aware catalogue. This is a method-specific improvement in the declared context screen, not a uniform complete-runtime advantage. All methods and both tasks remain in the supplementary validation table.

Cold-start work remains an exacting comparison. In the two-state task the median complete hat–ReLU process takes about 0.465 seconds, against 0.365 for adaptive search. In the eight-state task the respective medians are about 1.618 and 0.666 seconds; both use 2,594 policy-query evaluations in the displayed workload. These counts exclude separately reported training queries. Thus the certified objective is now an executed arm, but its execution is not an identified complete-work improvement. The recorded fitting, prediction and descendant counts locate the cost rather than attributing every successful return to a successful predictor.

### F.2. *Prospective validation and reuse*

Each fitted readout is frozen before a new predictor-independent context stream drawn uniformly from the sixteen-bit dyadic state grid is generated. There are 256 contexts per nonterminal date. The validation calculation executes the declared complete-action mesh recovery whenever the pre-query screen fails. Its clipped loss bounds the additional root queries on every recorded context. The supplementary table gives the achieved loss, actual recovery queries, simultaneous validation upper bound and descendant counts for every task, date, method and seed. These 6,144 method–context–date evaluations validate the stated context law and fallback; they are not relabeled as independent on-policy recursive states.

Each reuse process trains and serializes once and then serves sixty-four predetermined blocks of eight trajectories. At block index 16 it deliberately receives a stale acquisition identifier and uses the adaptive benchmark. Loading, hashing, identity validation, rejection, queries, cash-path enclosures, durable files and process shutdown are included. A stable observed crossing must begin by block index 48 and remain below the matched adaptive prefix through index 63. This requirement is fixed before execution.

TABLE F.I

SIXTY-FOUR-BLOCK PROSPECTIVE REUSE

| $(d, T)$ | Rule | Policy queries | Median (s) | Range (s) | Wins | Stable |
|---|---|---|---|---|---|---|
| 2,2 | A | 10057 | 3.621 | [3.169, 7.129] | – | – |
| 2,2 | B | 10392–10502 | 3.521 | [3.320, 3.572] | 4/8 | 4/8 |
| 2,2 | H | 10332–10447 | 3.746 | [3.320, 5.075] | 4/8 | 3/8 |
| 2,2 | Q | 9982–9992 | 3.621 | [3.120, 4.473] | 4/8 | 3/8 |
| 8,3 | A | 218656 | 24.914 | [24.762, 25.216] | – | – |
| 8,3 | B | 220282–220579 | 26.444 | [26.068, 26.921] | 0/8 | 0/8 |
| 8,3 | H | 220498–222019 | 26.469 | [25.915, 27.568] | 0/8 | 0/8 |
| 8,3 | Q | 223929–226193 | 27.872 | [26.964, 28.973] | 0/8 | 0/8 |

A complete-time win is relative to the adaptive process with the same repetition index. Stable means an observed cumulative crossing by index 48 that persists through index 63. The last summary and shutdown belong to complete clocks, not prefix clocks. The stale identity at index 16 is declined in every process. The same adaptive repetition is shared across matched seeds; these comparisons are not independent trials.

Three of the eight hat–ReLU processes in the two-state task satisfy that stable-prefix criterion; four have smaller final complete times than their matched adaptive process. The Bernstein and quadratic rules also exhibit small-task crossings. None of the learned processes wins the eight-state reuse comparison. In the small task, the adaptive complete times range from about 3.17 to 7.13 seconds, while hat–ReLU ranges from about 3.32 to 5.07 seconds. Consequently, a favorable matched crossing cannot be interpreted as uniform amortized superiority. The operation counts and every unfavorable repetition remain in the record.

The deployment theorem states what additional context-law and descendant-cost information would be sufficient for a prospective work bound. The experiment does not substitute its identity check for a distributional drift bound. Likewise, the sixty-four-block window is not extrapolated to an unobserved stopping horizon.

### F.3. *A cost comparison at the declared precision*

The central inference change is the Bellman-centered observation of Theorem 1. It estimates the actual expected cost difference by evaluating optimal advantages along each actual policy trajectory. The conditional Bellman expectations are evaluated with a tighter tolerance and paid for separately. The estimator is not the difference between two policy-loss upper bounds, and its realized observation is not asserted equal to a same-shock cash-cost difference.

For each scalar task the protocol fixes 1,024 new common-path rows, forty-eight-bit continuous bins and evaluation tolerance $\varepsilon/32$. The policies are adaptive search, certificate-aware hat–ReLU, certificate-aware Bernstein, and the original R67 ReLU controller at its pinned fitting seed. That historical controller is loaded unchanged. The predeclared loss margin is $1/256$, with numerical and sampling half-width allocations of $1/512$ each. No new row is added after inspecting a sign or a failed comparison. There are twelve paired contrasts across the two tasks on one centered-inference event with failure probability at most 0.01.

All twelve centered intervals lie within $[-1/256, 1/256]$. They therefore establish two-sided equivalence at the declared normalized-loss margin, including comparisons involving the unchanged historical neural controller. Every interval still contains zero: strict cost superiority is not identified. Both precision allocations are met. This replaces the earlier inability to assess the declared margin by a positive, original-law eligibility result, without changing the economic estimand or assuming a fitted value is exact.

For adaptive-minus-hat–ReLU, the two-state interval is approximately $[-0.00068263, 0.00070764]$ and the eight-state interval is approximately $[-0.00160291, 0.00160253]$, with outward display rounding. The corresponding cash-cost intervals on the same new rows have widths about 0.423 and 0.529. The centered widths are about 0.00139 and 0.00321. This is an identified precision improvement for the same expected-cost target, not an identified neural speed advantage. The complete shared inference processes cost about 9.03 and 408.63 seconds, respectively. Conditional Bellman integration, rather than state-bin width, dominates their numerical enclosure budget.

TABLE F.II

BELLMAN-CENTERED EXPECTED POLICY-COST DIFFERENCES

| $(d, T)$ | Pair | Simultaneous interval | Numerical width | Radius | Margin |
|---|---|---|---|---|---|
| 2,2 | A–B | [-0.000683, 0.000709] | 0.000194 | 0.000599 | Yes |
| 2,2 | A–H | [-0.000683, 0.000708] | 0.000194 | 0.000599 | Yes |
| 2,2 | A–R | [-0.000658, 0.000728] | 0.000188 | 0.000599 | Yes |
| 2,2 | B–R | [-0.000670, 0.000715] | 0.000187 | 0.000599 | Yes |
| 2,2 | H–B | [-0.000695, 0.000695] | 0.000193 | 0.000599 | Yes |
| 2,2 | H–R | [-0.000670, 0.000715] | 0.000187 | 0.000599 | Yes |
| 8,3 | A–B | [-0.001603, 0.001603] | 0.000812 | 0.001197 | Yes |
| 8,3 | A–H | [-0.001603, 0.001603] | 0.000812 | 0.001197 | Yes |
| 8,3 | A–R | [-0.001583, 0.001613] | 0.000801 | 0.001197 | Yes |
| 8,3 | B–R | [-0.001583, 0.001613] | 0.000802 | 0.001197 | Yes |
| 8,3 | H–B | [-0.001603, 0.001603] | 0.000812 | 0.001197 | Yes |
| 8,3 | H–R | [-0.001583, 0.001613] | 0.000802 | 0.001197 | Yes |

Pair first minus second; R is the unchanged R67 ReLU readout at seed 6603. Every task uses 1,024 new common-path rows, not that number per pair. All intervals are outward-rounded. Margin denotes containment in the prospectively specified interval $[-1/256, 1/256]$, not strict superiority. The numerical term is a full width; the sampling term is a half-width.

The two controllers' common initial-value terms cancel in expectation. All compatible true states within a sampled bin remain covered. No new inference trajectory in this catalogue has an unresolved acquisition branch. The raw conditional-value width, propagated bin width, acquisition allowance and final arithmetic diagnostics are reported separately; they are not assumed statistically independent, and pre-intersection component widths are not falsely presented as an additive decomposition of the tighter final interval.

### F.4. *Two controls and the economic service*

The vector study compares simplicial search, largest-edge uniform refinement, a trained two-output hat readout and a trained two-output quadratic proposal at two state dimensions and horizons two and three. Both learned rules purchase the

same twelve teacher states per nonterminal date. Their training objective is action-label loss, not the scalar convex certificate loss. All receive the same triangular lower certificate and feasible rounding rule under two independent continuous innovations. The horizon-four hat proposal transfers its three-date model and is identified as a transfer, not as a four-date fitted model.

Supplement Table L.II reports every two-control workload.

The vector records contain actual continuous-bin policy paths, complete clocks, peak resident memory and descendant counts. Eight paths per ordinary service, and one per horizon-four transfer service, are workload diagnostics rather than a precision-designed population comparison. The experiment adds a trained vector comparison, but it does not identify a vector expected-cost ranking or remove continuation-tree growth. Derivative branch-and-bound is not an executed arm of this catalogue.

For the scalar deployment decision, the release verifies that the cost-inference readouts coincide with the relevant cold and reuse readouts at seed 6901. Parameter matching is necessary for pairing these records. Under the additional same-numerical-policy premise, Proposition 9 gives their complete-service adoption region, with normal-policy multiplier 504 and observed work differences. Cross-platform prediction portability is separately discussed in the supplement. The supplementary table reports the work savings and critical price ratios, including cases in which savings are negative. The loss margin is a prospective numerical design choice, not an estimated installation charge. Equivalence at that margin makes policy quality an eligible comparison; adoption depends on declared prices and complete work, including newly purchased qualification.

### F.5. *Earlier evidence and independent reconstruction*

The original R67 catalogue remains a separate, unchanged companion, together with all preceding adverse studies. It records 212 returns, no neural complete-reuse winner, unresolved ordinary cost intervals, forty-eight neural fitting warnings and a severe horizon-four vector expansion. Those observations are not overwritten by the

new loss, new inference rows or different reuse window. The previous accumulation-aliasing counterexample and corrected execution also remain unchanged.

The new publication audit verifies every R69 receipt and stored-file identity, recomputes the achieved rational training certificates, checks every validation loss and executes independently written rational path calculations. A second interval implementation uses exact full-bin state and cost bounds, not only representative points, for unambiguous acquired paths. Its coverage and every diagnostic disagreement are recorded explicitly. This provides independent reconstruction of the sampled cash-path calculation and independent final inference arithmetic. It is not represented as an independent generation of every optimal-advantage endpoint or of the largest vector Bellman descendant tree.

## APPENDIX G: ACHIEVED READOUTS AND INDEPENDENT VALIDATION

The two certificate-aware feature families use the same purchased states, action labels, endpoint enclosures, penalty and iteration budget. The replay verifies equality of every training-array payload at a fixed task and seed before comparing the methods. The label-loss readout shares the hat features but not the objective. The quadratic arm shares information but has a different parameter count. A finite-dimensional hat function has both a conventional piecewise-linear description and an exact ReLU realization; changing the encoding does not create a new independent method.

TABLE G.I: Achieved convex readout certificates

| $(d,T,r)$ | Rule | Seed | Weights | $F(w)$ | $D(w)$ | $F(w)/\tau^2$ |
|---|---|---|---|---|---|---|
| 2,2,2 | B | 1 | 10 | 0.0187 | 1.72e-08 | 7.35e+04 |
| 2,2,2 | B | 2 | 10 | 0.0161 | 2.56e-08 | 6.33e+04 |
| 2,2,2 | H | 1 | 10 | 0.0183 | 9.8e-08 | 7.19e+04 |
| 2,2,2 | H | 2 | 10 | 0.0154 | 3.26e-08 | 6.05e+04 |
| 8,3,2 | B | 1 | 40 | 0.00909 | 2.47e-12 | 1.89e+04 |
| 8,3,3 | B | 1 | 40 | 0.0112 | 0 | 2.34e+04 |
| 8,3,2 | B | 2 | 40 | 0.00891 | 5.94e-10 | 1.85e+04 |
| 8,3,3 | B | 2 | 40 | 0.00883 | 1.09e-09 | 1.84e+04 |

| $(d,T,r)$ | Rule | Seed | Weights | $F(w)$ | $D(w)$ | $F(w)/\tau^2$ |
|---|---|---|---|---|---|---|
| 8,3,2 | H | 1 | 40 | 0.00908 | 8.07e-10 | 1.89e+04 |
| 8,3,3 | H | 1 | 40 | 0.0112 | 0 | 2.34e+04 |
| 8,3,2 | H | 2 | 40 | 0.00882 | 8.81e-10 | 1.83e+04 |
| 8,3,3 | H | 2 | 40 | 0.00877 | 4.43e-10 | 1.82e+04 |

These are achieved objective and optimization-gap values, not optimizer flags or hidden-layer convergence claims. The complete rational fractions are retained in DETAILS70.json. Repeated services of the same readout are listed once here and remain separately charged in their receipts.

The objective and dual gap are evaluated on the returned quantized weights using rational contexts and features. The optimization certificate remains valid whether an approximate optimization routine reports success or a warning. The table concerns the achieved convex box problem, not unrestricted hidden-layer optimization. Raw weights, endpoints, labels and exact fractions are retained. Previously reported neural-fitting warnings belong to the unchanged earlier catalogue and are not removed by the successful computation of a new objective.

*Exact context law and counting.* Training and validation coordinates are uniform on the sixteen-bit dyadic grid, not on the continuous state cube or the full twenty-four-bit acquisition grid. Matching methods share these validation draws. The 6,144 method–context–date evaluations are not pooled as 6,144 independent states: every stated validation bound uses its own 256-row record. The separate cost-inference experiment retains the original continuous initial and innovation laws.

TABLE G.II: Independent-context work-loss validation

| $(d,T,r)$ | Rule | Seed | Returns | Root mean | Upper | All recovery queries |
|---|---|---|---|---|---|---|
| 2,2,2 | H | 1 | 48 | 13.81 | 16.442 | 17680 |
| 2,2,2 | H | 2 | 50 | 13.68 | 16.315 | 17510 |
| 2,2,2 | B | 1 | 25 | 15.34 | 17.000 | 19635 |
| 2,2,2 | B | 2 | 21 | 15.61 | 17.000 | 19975 |
| 2,2,2 | L | 1 | 42 | 14.21 | 16.973 | 18190 |
| 2,2,2 | L | 2 | 47 | 13.88 | 16.482 | 17765 |

| $(d,T,r)$ | Rule | Seed | Returns | Root mean | Upper | All recovery queries |
|---|---|---|---|---|---|---|
| 2,2,2 | Q | 1 | 45 | 14.01 | 16.707 | 17935 |
| 2,2,2 | Q | 2 | 52 | 13.55 | 16.282 | 17340 |
| 8,3,2 | H | 1 | 21 | 15.61 | 17.000 | 19975 |
| 8,3,3 | H | 1 | 23 | 15.47 | 17.000 | 462088 |
| 8,3,2 | H | 2 | 13 | 16.14 | 17.000 | 20655 |
| 8,3,3 | H | 2 | 21 | 15.61 | 17.000 | 469565 |
| 8,3,2 | B | 1 | 25 | 15.34 | 17.000 | 19635 |
| 8,3,3 | B | 1 | 23 | 15.47 | 17.000 | 462088 |
| 8,3,2 | B | 2 | 11 | 16.27 | 17.000 | 20825 |
| 8,3,3 | B | 2 | 20 | 15.67 | 17.000 | 471130 |
| 8,3,2 | L | 1 | 9 | 16.40 | 17.000 | 20995 |
| 8,3,3 | L | 1 | 6 | 16.60 | 17.000 | 486452 |
| 8,3,2 | L | 2 | 5 | 16.67 | 17.000 | 21335 |
| 8,3,3 | L | 2 | 13 | 16.14 | 17.000 | 481419 |
| 8,3,2 | Q | 1 | 8 | 16.47 | 17.000 | 21080 |
| 8,3,3 | Q | 1 | 21 | 15.61 | 17.000 | 464894 |
| 8,3,2 | Q | 2 | 10 | 16.34 | 17.000 | 20910 |
| 8,3,3 | Q | 2 | 4 | 16.73 | 17.000 | 493938 |

Each row has 256 new contexts; seed suffix 1 or 2 denotes 6901 or 6902. The upper bound is independently recomputed from an exact sample sum and an outward radius. It applies to the declared complete-action mesh recovery and context law, not automatically to an adaptive on-policy stream. Recovery-query totals include descendants.

The validation family contains twenty-four task–date–method–seed records with 256 contexts per record. The logarithm allowance twelve covers the stated finite family by a union bound. Reusing a context stream across methods induces dependence between estimators but does not invalidate that bound. The same stream is not counted as additional independent information for each method. The deterministic work inequality is checked on every context. The distributional upper endpoint is recomputed from the exact sample sum, an eighty-digit square root with an explicit expansion, and outward conversion to the printed numerical endpoint.

The full-action recovery mesh in this validation design has a declared root cap $K$. Its actual descendant queries are separately recorded. These observations do not validate a different adaptive fallback by definition. In Theorem 8, a distributional transfer allowance and a complete endpoint-work cap are additional premises. The theorem therefore handles predictable choices and dependent contexts without asserting that the recorded recursive state stream is independent.

## APPENDIX H: PRECISION AND THE CONTINUOUS-LAW ESTIMAND

TABLE H.I

PAID INFERENCE WORK AND RAW ENCLOSURE COMPONENTS

| $(d,T)$ | Rule | Evaluator queries | Arm (s) | Conditional | State-bin | Ambiguous |
|---|---|---|---|---|---|---|
| 2,2 | A | 101226 | 1.98 | 0.000113 | 2.3e-13 | 0 |
| 2,2 | B | 101226 | 1.91 | 0.000113 | 2.3e-13 | 0 |
| 2,2 | H | 101226 | 1.91 | 0.000113 | 2.3e-13 | 0 |
| 2,2 | R | 101226 | 1.83 | 0.000113 | 2.3e-13 | 0 |
| 8,3 | A | 2921243 | 100.74 | 0.000595 | 3.7e-13 | 0 |
| 8,3 | B | 2921234 | 102.56 | 0.000596 | 3.7e-13 | 0 |
| 8,3 | H | 2921234 | 102.59 | 0.000596 | 3.7e-13 | 0 |
| 8,3 | R | 2921465 | 101.36 | 0.000597 | 3.7e-13 | 0 |

Component columns are mean full widths before valid regret-range intersections. They do not add to the final narrower contrast width. Arm clocks are internal diagnostics; complete shared inference is charged as a whole: 9.03 seconds, 408.63 seconds. State-bin widths include propagated arithmetic. No acquisition ambiguity is discarded.

The centered observation is the sum of optimal Bellman advantages along an actual policy trajectory. Its expectation equals policy loss. Subtracting two such observations cancels the common initial optimal value in expectation. It does not cancel cash-flow risk path by path, and no empirical fitted value is substituted for the optimal value. Each conditional Bellman evaluation uses its own tolerance, action covering and continuous-innovation integration account, whose work is charged to the inference service.

The twelve paired intervals form one finite family over the two declared scalar tasks. With logarithm allowance twelve, $2 \cdot 12 \exp(-12) < 0.01$. For example, a rational partial sum of the exponential series already proves $\exp(12) > 2400$; no rounded logarithm is needed for the coverage claim. The ordinary cash-cost estimator uses a separate empirical-Bernstein family on the same rows. A joint claim about both families requires the sum of their tail budgets, not an assumption that the estimators are independent.

The mean conditional-value and state-bin component widths are recorded before intersecting with valid local regret ranges. The intersection can substantially reduce the resulting interval. Accordingly, those diagnostic widths need not sum to the final centered width, and they are not interpreted as independent errors. The state-bin column includes propagated state arithmetic. The actual final interval width is the mean final numerical width plus the two sampling radii, before any final support clipping.

For every path, the numerical action used by the controller is fixed before the more accurate evaluation. Its feasibility is checked on the true-state box as well as on the acquired state. An unresolved acquisition branch must receive a valid remaining support; it cannot be discarded. None is encountered in the new inference catalogue. The absence of sampled ambiguity is an observation, not a theorem excluding acquisition boundaries from the model.

## APPENDIX I: INDEPENDENT FULL-BIN RECONSTRUCTION

The new independent implementation imports no NBO numerical kernel. It uses exact rational intervals for the continuous initial and innovation bins, twenty-four-bit acquired-state comparisons, true-state feasibility, stage costs, transitions and terminal costs. It treats the two cumulative endpoints as separate immutable rational numbers. In a transition coordinate, the deterministic function

$$f(u,v) = \frac{1}{16} + \frac{u}{2} + \frac{v}{8} + \frac{u(1-v)}{16}$$

has derivatives $1/2 + (1-v)/16 \geq 0$ and $1/8 - u/16 \geq 0$ on the state cube. Thus evaluation at the lower–lower and upper–upper corners bounds its range without

invoking the production interval library. Quadratic costs use exact interval squares; shortage uses the monotone positive-part map followed by an interval square. Both shocks keep their original widths and marginal laws.

The implementation checks that its complete bin enclosure lies inside the archived cash-cost enclosure. A second, independently written rational midpoint calculation checks acquisition and cumulative cost at a represented point. The midpoint test is not substituted for the full-bin test. Across 3,706 physical path archives there are 38,722 recorded path rows; 736 distinct payloads contain 14,242 unique rows. Repeated timing copies are hash-checked and compared, but their duplicate rational evaluation is not counted again. The independent full-bin replay covers all 14,242 unique rows, with no unresolved branch in this executed catalogue.

A separate exact-sum and high-precision calculation checks every final centered confidence interval from its stored advantage endpoints. This independently verifies final inference arithmetic, not the generation of those optimal-advantage endpoints. The regression suite compares primitive identities and small complete value problems with separate rational formulas. It does not independently regenerate the largest multivariate Bellman descendant tree. The release states these different coverage levels rather than assigning an undifferentiated independent-verification label to the whole program.

### I.1. *Prediction portability and retained disagreements*

An initial strict replay stopped because one Bernstein validation prediction was not bitwise equal to a recomputed value. That failed attempt, its original audit source and traceback are retained on their own branch and in the publication inputs. The failure does not by itself identify an invalid Bellman bound. Nor does a successful rerun on another processor prove the earlier numerical result bitwise reproducible.

The current audit checks the saved actual action on its declared lattice and recomputes its strong-convexity chord with exact rational arithmetic. A discrepancy with a recomputed readout is additionally checked against the declared action-rounding and prediction allowances, not accepted by an arbitrary numerical closeness thresh-

old. The original scientific endpoints, actions, models and source freeze are unchanged. Numerical parameter identity, callable-policy identity and successful certificate verification are distinct statements.

In particular, equality of saved readout arrays is necessary but not sufficient for transferring a cost estimate to an implementation on a different arithmetic runtime. The complete-service purchase region assumes the same numerical callable policy or an additional valid implementation-transfer bound. Matching processor metadata or passing a finite replay is not silently promoted to an all-state policy-identity theorem.

## APPENDIX J: COMPLETE WORK AND CONDITIONAL ADOPTION

TABLE J.I: Conditional reuse-stage adoption thresholds

| $(d,T)$ | Rule | Repeat | Saving (s) | Loss upper | $p/v$ threshold |
|---|---|---|---|---|---|
| 2,2 | H | 0 | 0.000 | 0.3441 | 1383.7120 |
| 2,2 | H | 1 | 0.350 | 0.3441 | 0.9821 |
| 2,2 | H | 2 | -0.201 | 0.3441 | – |
| 2,2 | H | 3 | 3.457 | 0.3441 | 0.0996 |
| 2,2 | B | 0 | -0.201 | 0.3440 | – |
| 2,2 | B | 1 | 0.601 | 0.3440 | 0.5727 |
| 2,2 | B | 2 | -0.401 | 0.3440 | – |
| 2,2 | B | 3 | 3.808 | 0.3440 | 0.0904 |
| 8,3 | H | 0 | -1.153 | 0.8079 | – |
| 8,3 | H | 1 | -1.301 | 0.8079 | – |
| 8,3 | H | 2 | -1.399 | 0.8079 | – |
| 8,3 | H | 3 | -1.555 | 0.8079 | – |
| 8,3 | B | 0 | -1.306 | 0.8079 | – |
| 8,3 | B | 1 | -1.606 | 0.8079 | – |
| 8,3 | B | 2 | -1.302 | 0.8079 | – |
| 8,3 | B | 3 | -1.955 | 0.8079 | – |

Seed 6901; zero incremental installation fee and common or previously paid qualification. The loss upper is 504 times the corresponding policy-cost upper endpoint. A displayed positive-saving threshold describes the adoption region only when the numerical callable policy is the one evaluated by inference. Saved parameter equality is checked, but cross-platform numerical equality is an additional requirement. A dash means no positive work saving in that matched observation. Newly purchased qualification work must be added before claiming a first-purchase saving. Prices are declared inputs, not estimated currency values.

*Qualification is not free.* The displayed reuse thresholds condition on a common or previously paid qualification calculation. The process receipts cover the complete reuse service, not the separate policy-cost inference experiment. For a first purchase, let $C_q$ be additional qualification work not paid by the benchmark alternative. In Proposition 9, use $W_\pi = W_\pi^{\mathrm{run}} + C_q$ and $W_\nu = W_\nu^{\mathrm{run}}$; the observed saving becomes $S - C_q$. This is a change of the explicit accounting boundary, not an additional theorem assumption about free inference. Charging the entire observed shared inference process gives $C_q$ approximately 9.03 seconds in the small task and 408.63 seconds in the larger task. Every displayed matched saving then becomes negative. Thus the current receipts do not identify a positive qualification-plus-reuse work saving under that conservative first-purchase boundary. An allocated marginal inference time or a longer repayment horizon was not separately executed. The machine-readable table retains both boundaries, and the general price region remains valid with the actual charged work and installation fee.

The multiplier 504 follows from sixty-four blocks of eight trajectories with one predetermined fallback block. In that block the service uses the benchmark itself, giving zero expected policy-cost difference. In the other blocks it uses the fixed candidate. The cost interval can therefore be scaled by 504 under the policy-identity premise of Proposition 9. A positive work saving is an observed matched process difference, not a confidence bound for future latency. The complete timing range also records all unfavorable repetitions and does not authorize selection of a favorable timing result as a population estimate.

The parent receipt includes child launch, imports, fitting or loading, all policy and path calculations, serialization, child exit and log flushing. Parent receipt

bookkeeping belongs to the group clock. Training, inference and publication replay have different boundaries and are not assigned zero cost when excluded from a displayed sub-account. The machine record includes the permitted CPU, thread counts, available frequency and governor data. Processor frequency was not locked by the experiment. Fine timing differences consequently remain descriptive even when operation counts reproduce exactly.

The original normalized investment units are retained. The loss margin is fixed by the numerical protocol; the value of a loss unit, compute price and installation fee are purchaser inputs. This produces a transparent conditional adoption region, not an estimated monetary welfare result. Its noninferiority and equivalence statements concern the actual expected-cost target and do not imply strict neural superiority.

## APPENDIX K: PRESERVATION AND SCOPE

Every active R67 theorem statement is retained verbatim in the current theoretical sections, and every proof is retained in the article or this supplement. The original empirical article and its full supplement are unchanged companion documents. They retain all earlier conventional winners, unresolved direct-cost comparisons, fitting warnings, vector horizon costs and the accumulation-aliasing counterexample. The broader NBO development remains available under its original diffusion, recursive-preference, temporal-self and game hypotheses; those hypotheses are not replaced by the convex expected-cost premises of the current query theorem.

The new frozen R69 catalogue is integrated once. Publication replay, repeated clocks, repeated random streams, duplicate path payloads and regenerated tables do not increase its number of independent economic observations. The final source and content map records the location and hash of each retained item. Compilation, regression tests and independent numerical calculations are evidence for their stated checks, not an external editorial decision or a formal proof-checker certificate.

TABLE L.I

COMPLETE COLD-START SERVICES AT A COMMON INFORMATION BUDGET

| $(d, T)$ | Rule | Processes | Policy queries | Median (s) | Range (s) |
|---|---|---|---|---|---|
| 2,2 | A | 4 | 597 | 0.365 | [0.364, 0.366] |
| 2,2 | B | 8 | 607–612 | 0.465 | [0.465, 0.466] |
| 2,2 | H | 8 | 592–612 | 0.465 | [0.465, 0.466] |
| 2,2 | L | 8 | 592–597 | 0.440 | [0.414, 0.466] |
| 2,2 | Q | 8 | 597 | 0.415 | [0.415, 0.967] |
| 2,2 | U | 4 | 597 | 0.390 | [0.364, 0.415] |
| 8,3 | A | 4 | 2594 | 0.666 | [0.615, 0.666] |
| 8,3 | B | 8 | 2594 | 1.618 | [1.618, 1.619] |
| 8,3 | H | 8 | 2594 | 1.618 | [1.618, 1.669] |
| 8,3 | L | 8 | 2594 | 1.518 | [1.517, 1.568] |
| 8,3 | Q | 8 | 2594 | 1.518 | [1.467, 1.519] |
| 8,3 | U | 4 | 2594 | 0.641 | [0.615, 0.666] |

A: adaptive; U: upper-chord minimizer; H: certificate-aware hat–ReLU; B: certificate-aware Bernstein; L: label-loss hat–ReLU; Q: quadratic. Queries concern the policy workload; training and validation queries are additional. Four process repetitions per seed are descriptive clocks, not independent fits or confidence bounds. All complete times include launch and shutdown.

## APPENDIX L: COMPLETE COLD-START AND VECTOR WORKLOADS

The tables retain every declared method and task. They are relocated intact from the article to keep its reading path focused; no row, adverse comparison or timing repetition is removed.

## APPENDIX M: NEW REUSE RECORDS AND INDEPENDENT ARITHMETIC

The new scientific source commitment binds all five new scientific modules and their inherited Python dependencies, together with the exact task and timeout catalogue. It precedes execution. The numerical study contains no newly fitted predictor and no new independent policy-cost sample. Its roots are predetermined localized states, not a uniform or stationary deployment distribution.

TABLE L.II

TWO-CONTROL AND TWO-INNOVATION COMPLETE SERVICES

| $(d, T)$ | Rule | Processes | Queries | Median (s) | Peak KiB |
|---|---|---|---|---|---|
| 2,2 | H | 2 | 98 | 0.47 | 65520 |
| 2,2 | Q | 2 | 99 | 0.49 | 65524 |
| 2,2 | S | 2 | 82 | 0.39 | 65164 |
| 2,2 | U | 2 | 82 | 0.36 | 65328 |
| 2,3 | H | 2 | 5634 | 5.83 | 66004 |
| 2,3 | Q | 2 | 5914 | 5.93 | 66256 |
| 2,3 | S | 2 | 4541 | 3.70 | 65184 |
| 2,3 | U | 2 | 9039 | 7.93 | 65016 |
| 2,4 | H | 1 | 872033 | 62.01 | 65872 |
| 2,4 | S | 1 | 592658 | 41.37 | 65256 |
| 8,2 | H | 2 | 91 | 0.46 | 65780 |
| 8,2 | Q | 2 | 91 | 0.46 | 65864 |
| 8,2 | S | 2 | 75 | 0.36 | 65252 |
| 8,2 | U | 2 | 75 | 0.36 | 65336 |
| 8,3 | H | 2 | 3495 | 5.93 | 66028 |
| 8,3 | Q | 2 | 3495 | 5.85 | 66208 |
| 8,3 | S | 2 | 2645 | 3.55 | 65168 |
| 8,3 | U | 2 | 3278 | 4.60 | 65544 |

S: simplicial; U: largest-edge uniform refinement; H and Q: trained two-output hat and quadratic proposals. Ordinary services have eight continuous-bin paths. Horizon-four services have one path, and H transfers its three-date readout. These are workload records, not precision-designed estimates of population policy costs. Peak memory is the maximum observed resident set within the listed processes.

TABLE M.I: Every new complete-process repetition

| Task | Rule | Rep. | Seconds | Queries | Hits | RSS |
|---|---|---|---|---|---|---|
| S2/2 | Fresh | 1 | 1.166251 | 490 | 0 | 97,804 |
| S2/2 | Exact | 1 | 0.715207 | 115 | 25 | 97,944 |
| S2/2 | Geometric | 1 | 0.665186 | 55 | 29 | 97,808 |
| S2/2 | Exact | 2 | 0.765338 | 115 | 25 | 97,816 |

| Task | Rule | Rep. | Seconds | Queries | Hits | RSS |
|---|---|---|---|---|---|---|
| S2/2 | Geometric | 2 | 0.668814 | 55 | 29 | 97,816 |
| S2/2 | Fresh | 2 | 1.216814 | 490 | 0 | 97,804 |
| S2/2 | Geometric | 3 | 0.668796 | 55 | 29 | 97,800 |
| S2/2 | Fresh | 3 | 1.115930 | 490 | 0 | 97,928 |
| S2/2 | Exact | 3 | 0.765269 | 115 | 25 | 97,932 |
| S8/3 | Fresh | 1 | 2.739717 | 1,610 | 0 | 97,936 |
| S8/3 | Exact | 1 | 1.921036 | 959 | 7 | 97,944 |
| S8/3 | Geometric | 1 | 1.115856 | 401 | 13 | 97,936 |
| S8/3 | Exact | 2 | 1.867363 | 959 | 7 | 97,932 |
| S8/3 | Geometric | 2 | 1.216160 | 401 | 13 | 97,936 |
| S8/3 | Fresh | 2 | 2.772324 | 1,610 | 0 | 97,932 |
| S8/3 | Geometric | 3 | 1.166799 | 401 | 13 | 97,932 |
| S8/3 | Fresh | 3 | 2.721673 | 1,610 | 0 | 97,944 |
| S8/3 | Exact | 3 | 1.867344 | 959 | 7 | 97,932 |
| V2/2 | Fresh | 1 | 2.422182 | 397 | 0 | 97,932 |
| V2/2 | Exact | 1 | 1.219324 | 125 | 17 | 97,932 |
| V2/2 | Geometric | 1 | 0.915594 | 61 | 21 | 97,932 |
| V2/2 | Exact | 2 | 1.216086 | 125 | 17 | 97,944 |
| V2/2 | Geometric | 2 | 0.919073 | 61 | 21 | 97,936 |
| V2/2 | Fresh | 2 | 2.470714 | 397 | 0 | 97,936 |
| V2/2 | Geometric | 3 | 0.915826 | 61 | 21 | 97,928 |
| V2/2 | Fresh | 3 | 2.471993 | 397 | 0 | 97,932 |
| V2/2 | Exact | 3 | 1.216277 | 125 | 17 | 97,936 |
| V2/3 | Fresh | 1 | 21.474661 | 25,534 | 0 | 98,072 |
| V2/3 | Exact | 1 | 14.203954 | 17,023 | 3 | 98,076 |
| V2/3 | Geometric | 1 | 9.388726 | 11,349 | 5 | 98,056 |
| V2/3 | Exact | 2 | 14.556513 | 17,023 | 3 | 98,056 |
| V2/3 | Geometric | 2 | 9.793062 | 11,349 | 5 | 98,008 |
| V2/3 | Fresh | 2 | 22.620244 | 25,534 | 0 | 98,064 |
| V2/4 | Fresh | 1 | 347.971704 | 5,085,947 | 0 | 99,212 |
| V2/4 | Exact | 1 | 168.526652 | 2,542,973 | 2 | 99,216 |
| V2/4 | Geometric | 1 | 84.068560 | 1,271,487 | 3 | 99,212 |

Each row is one separately launched and completed process. No clock repetition is a new fitting seed or an independent policy-cost observation. All stage and source identities remain in its receipt.

TABLE M.II: Paid fills, rejected records and lookup overhead

| Task | Rule | Rep. | Checks | Stored | Stale | Fill (s) | Lookup (s) |
|---|---|---|---|---|---|---|---|
| S2/2 | Fresh | 1 | 0 | 0 | 0 | 0.5554 | 0.0000 |
| S2/2 | Exact | 1 | 25 | 4 | 3 | 0.1301 | 0.0055 |
| S2/2 | Geometric | 1 | 30 | 2 | 1 | 0.0642 | 0.0039 |
| S2/2 | Exact | 2 | 25 | 4 | 3 | 0.1349 | 0.0056 |
| S2/2 | Geometric | 2 | 30 | 2 | 1 | 0.0641 | 0.0040 |
| S2/2 | Fresh | 2 | 0 | 0 | 0 | 0.5678 | 0.0001 |
| S2/2 | Geometric | 3 | 30 | 2 | 1 | 0.0621 | 0.0041 |
| S2/2 | Fresh | 3 | 0 | 0 | 0 | 0.5363 | 0.0000 |
| S2/2 | Exact | 3 | 25 | 4 | 3 | 0.1344 | 0.0051 |
| S8/3 | Fresh | 1 | 0 | 0 | 0 | 2.1140 | 0.0000 |
| S8/3 | Exact | 1 | 7 | 5 | 4 | 1.2850 | 0.0036 |
| S8/3 | Geometric | 1 | 14 | 2 | 1 | 0.5343 | 0.0029 |
| S8/3 | Exact | 2 | 7 | 5 | 4 | 1.2600 | 0.0039 |
| S8/3 | Geometric | 2 | 14 | 2 | 1 | 0.5502 | 0.0029 |
| S8/3 | Fresh | 2 | 0 | 0 | 0 | 2.1810 | 0.0000 |
| S8/3 | Geometric | 3 | 14 | 2 | 1 | 0.5306 | 0.0033 |
| S8/3 | Fresh | 3 | 0 | 0 | 0 | 2.1113 | 0.0000 |
| S8/3 | Exact | 3 | 7 | 5 | 4 | 1.2652 | 0.0038 |
| V2/2 | Fresh | 1 | 0 | 0 | 0 | 1.8126 | 0.0001 |
| V2/2 | Exact | 1 | 17 | 4 | 3 | 0.5964 | 0.0040 |
| V2/2 | Geometric | 1 | 22 | 2 | 1 | 0.3040 | 0.0032 |
| V2/2 | Exact | 2 | 17 | 4 | 3 | 0.6196 | 0.0042 |
| V2/2 | Geometric | 2 | 22 | 2 | 1 | 0.3143 | 0.0033 |
| V2/2 | Fresh | 2 | 0 | 0 | 0 | 1.8519 | 0.0000 |
| V2/2 | Geometric | 3 | 22 | 2 | 1 | 0.3170 | 0.0033 |
| V2/2 | Fresh | 3 | 0 | 0 | 0 | 1.8495 | 0.0000 |
| V2/2 | Exact | 3 | 17 | 4 | 3 | 0.6215 | 0.0041 |
| V2/3 | Fresh | 1 | 0 | 0 | 0 | 20.8660 | 0.0000 |

| Task | Rule | Rep. | Checks | Stored | Stale | Fill (s) | Lookup (s) |
|---|---|---|---|---|---|---|---|
| V2/3 | Exact | 1 | 3 | 3 | 2 | 13.5845 | 0.0009 |
| V2/3 | Geometric | 1 | 6 | 2 | 1 | 8.7833 | 0.0013 |
| V2/3 | Exact | 2 | 3 | 3 | 2 | 13.9392 | 0.0011 |
| V2/3 | Geometric | 2 | 6 | 2 | 1 | 9.1646 | 0.0012 |
| V2/3 | Fresh | 2 | 0 | 0 | 0 | 21.9832 | 0.0000 |
| V2/4 | Fresh | 1 | 0 | 0 | 0 | 347.3480 | 0.0000 |
| V2/4 | Exact | 1 | 2 | 2 | 0 | 167.8846 | 0.0006 |
| V2/4 | Geometric | 1 | 3 | 1 | 0 | 83.4527 | 0.0006 |

Stored is the maximum number of retained witnesses. Stale counts discarded records at the declared context invalidation. Fill and lookup are internal subclocks, not the complete process boundary. The lookup account includes record-load validation. Repeated storage, hashing, output and shutdown are additionally included in complete receipts.

### M.1. *Complete timing and retained records*

The process receipt includes creation, Python imports, original Bellman fills, all recursive descendants, cache retrieval, rational checks, repeated serialization, final output and process exit. The driver writes a separate receipt after that boundary. Summing child block clocks is not the complete process time. Every block's body prefix excludes imports and has its own exact definition in the source. The complete receipts, rather than those prefixes, determine the reported timing comparison.

No processor frequency lock is claimed. The start and end records include the selected CPU, frequency fields where the environment exposes them, interpreter version and platform. A cyclic mode order limits a simple fixed-order confound but does not create randomized hardware-frequency inference. Close clock differences require more evidence than the prescribed finite repetitions. Bellman action counts are consequently reported separately from seconds. They are invariant descriptive counts for these transcripts, not predictions of another implementation's throughput.

The 128-record cap bounds logical retained witnesses. It does not bound total process resident memory: a fresh recursive fill can retain batched descendants,

interpreter state, interval work arrays and output buffers. The largest process memory remains in the table. The executed cache operates at workload roots only. The four-date pressure is not hidden by counting a later root hit as though the first fill used no descendants.

### M.2. *Independent checks and trusted premises*

The additional admission audit matches all 708 returns to the prescribed state, control dimension, horizon, tolerance and context epoch. Cache reloads in this catalogue use the same process and numerical model. Arbitrarily supplied external intervals are not made valid by deserialization or a file hash.

For a returned hit, the independent checker locates the earlier paid source record with the same context, horizon, source state, action and endpoints. It recomputes the original regularity constants, state distance, action displacement and interval expansion in exact rational arithmetic. It checks the returned action's lattice and capacity and that the serialized binary endpoints contain the exact expanded interval. A narrow stored interval alone would not prove that a source fill encloses the original value: the inherited verifier and its primitive proof remain explicit premises.

For conditional ReLU integration, the second implementation reconstructs the original projected drift and shocks independently. It does not call the divided-power formula. It clips the support rectangle against the affine neuron's positive half-plane and uses oriented area and first moments. Degenerate projections reduce to one-dimensional integration or a deterministic positive part. All rational discrepancies are fatal to the new arithmetic audit and are retained as evidence if encountered.

The earlier R70 independent cash-path audit and final inference arithmetic are unchanged. The historical full descendant trees are not certified by pretending that a file hash is a mathematical proof. Neither the new rational transfer checks nor the polygon integrations independently regenerate all those trees. The release states this boundary in its machine-readable audit and response.

### M.3. *Development probe and publication preservation*

Before the new source commitment, a vector development probe queried $(0.4, 0.6)$ at tolerance 0.04 and horizons two, three and four. The first two returned. The probe was stopped at its 35-second process limit while computing horizon four. Its log is retained separately and is not merged with the later 900-second scientific catalogue or used as a favorable timing observation.

Every R70 active section is assigned to exactly one current document and checked byte for byte. All inherited formal statements and proofs are thereby retained, not reconstructed from a favorable summary. The complete old main article and supplement remain companion PDFs. New tables are derived from the committed receipts and independent audit, and all four current documents are rebuilt from a clean source directory. Visual inspection is documented against the exact PDF hashes actually inspected. Compilation and tests do not constitute external mathematical acceptance.

## APPENDIX N: COMPLETE CONTINUATION-QUALIFICATION RECORDS

The fixed design was committed before any new construction service ran. Its four points are the product of covers with $8^2$ or $16^2$ state nodes and two nine-coefficient bases. The terminal continuation is the original $g$; each preceding date is fitted against the continuation actually returned at the following date. The stage label is an interval containing the full continuous-action Bellman infimum. The coefficient box, quantum, feature knots, action-cover size and optimization tolerance were fixed in advance.

The ReLU basis consists of the constant and, in each coordinate, the identity and positive parts at $1/4$, $1/2$ and $3/4$. The conventional basis consists of the constant and the four nonconstant degree-four Bernstein polynomials per coordinate. Their valid feature moduli are respectively one and four. Having the same number of coefficients does not make their function classes or approximation moduli identical. Each basis receives its own actual fixed-future labels and the same construction rule; the label-loss ablation within each stage shares those exact labels and features.

TABLE N.I

EVERY ATTAINED READOUT AND ITS FIXED-LABEL LEAST-SQUARES ABLATION

| $n$ | Basis | Date | $\mathcal{W}$ upper | LS bound upper | Dual gap |
|---|---|---|---|---|---|
| 8 | ReLU | 0 | 2.38478 | 3.59860 | 0.000000007 |
| 8 | ReLU | 1 | 3.21372 | 4.91701 | 0.000000014 |
| 8 | Bernstein | 0 | 1.92163 | 5.25637 | 0.000000000 |
| 8 | Bernstein | 1 | 3.67363 | 9.63856 | 0.000000000 |
| 16 | ReLU | 0 | 1.63091 | 1.99849 | 0.000000014 |
| 16 | ReLU | 1 | 2.24222 | 2.75521 | 0.000000014 |
| 16 | Bernstein | 0 | 1.59492 | 2.78996 | 0.000000000 |
| 16 | Bernstein | 1 | 3.02915 | 5.10081 | 0.000000000 |

LS is bounded least squares on the identical features and exact fixed-future interval midpoints. Its bound is evaluated by the same residual-qualification formula. It is not a separately propagated controller. The primal and dual calculations are exact rational checks, independently reconstructed without relying on the optimizer status.

Every linear-program output is rounded into its declared coefficient box and checked with rational probability multipliers. The table's gap is the achieved primal value minus the exact dual lower bound, not the solver's reported duality tolerance. Constant recentering changes the continuation gauge without changing this objective. The predeclared success tolerance is $1/65536$. The stored status and message remain even when the exact certificate succeeds; an unsuccessful optimizer would retain its feasible fallback and its actual gap rather than disappear from the catalogue.

The service clock includes both backward constructions, all exact action and feature-mean evaluations, the qualification optimizer, its rationalization, the matched bounded least-squares ablation and the service output. The separate clock record identifies its endpoint. These are single recorded construction times; no estimate of mean runtime, clock stability or a universal neural speed ratio is inferred. The independent audit has its own time and never replaces a scientific service time. Exact arithmetic-operation counts are not a bound on rational bit complexity.

TABLE N.II

COMPLETE NEW CONSTRUCTION WORK AND OBSERVATION SCOPE

| $n$ | Basis | Action values | Feature means | Seconds | Cost rows |
|---|---|---|---|---|---|
| 8 | ReLU | 8448 | 38016 | 1.174 | 0 |
| 8 | Bernstein | 8448 | 38016 | 3.174 | 0 |
| 16 | ReLU | 33792 | 152064 | 4.793 | 0 |
| 16 | Bernstein | 33792 | 152064 | 10.617 | 0 |

Action values include continuous-action covering and whole-cell feasible proposals at both dates. Feature means count evaluations for the fitted future, not terminal formulas. Seconds include both fits, qualification, matched least-squares ablations and the service output; the independent checker has a separate clock. There are no recursive optimal-value queries and no new independent policy-cost observations. Single-service times are descriptive.

The independent action-value evaluator does not invoke the production positive-part moment expansion. It partitions the innovation interval at every ReLU breakpoint and integrates affine pieces. For the degree-four conventional continuation it uses the positive Boole weights on five equally spaced points; the result is exact for this polynomial degree. Final-date values are checked against the retained original rational investment primitive. It also reconstructs every feasible node action, the original continuous-action lower enclosure, the robust acquired action and its whole-cell gap. Thus the newly computed residual account is tied to original primitives rather than only to a stored hash.

The current construction does not sample a new expected policy-cost difference. The support supplied by its fitted continuation is available to Theorem 15, but no new stopped investment comparison is asserted. The separately executed fixed-sample comparisons, two-control workloads and cache processes retain their original observations, contracts and coverage accounts.

## APPENDIX O: EXACT SEQUENTIAL-LEDGER REGRESSIONS

The new regression suite checks conditional means, range reservation, target changes and failed numerical episodes. Its adaptive binary-tree diagnostic enumerates all $2^{12} = 4096$ leaves and every prefix, with exact probabilities and changing

predictable ranges. The event count is a finite-model check of the implemented boundary, not an empirical substitute for the supermartingale proof. Rational logarithm bounds and coefficient duals are also tested independently of floating solver stopping flags.

A support failure and a numerical failure are deliberately different branches. An empty intersection or a changed fixed target is rejected. A numerical failure records the prequalified full support, advances the prefix and charges its work. The interface cannot prove that a caller fixed the support before observing the episode; that temporal and probabilistic premise remains part of the algorithm's contract. A cache carried across episodes is included in the target definition or is handled as varying conditional cost.

All prior scientific arrays and sources remain unchanged. The ordinary-source build runs the inherited and current tests and recomputes the published tables. The clean archive includes the transitive R62 input sources, including the original service kernel omitted by the interrupted R72 archive. The reproduction report identifies the suites run inside that archive and compares actual PDF text and page counts after deleting generated PDFs and tables. Visual inspection is recorded separately against the output hashes. None of those publication checks is an external mathematical acceptance decision.

## APPENDIX: REFERENCES